

Primes in the Wires: Spectral Correspondence Between a Quantum Graph and the Riemann Zeros

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ABSTRACT

We report a computational search over quantum graph space for a realization of the Hilbert-Pólya conjecture — the proposal that the nontrivial zeros of the Riemann zeta function are eigenvalues of a self-adjoint operator. Exhaustive search over all 853 connected 7-vertex topologies identifies the cycle graph C_7 as optimal. Under standard Neumann vertex conditions, spectral correspondence plateaus at a topology- and size-independent ceiling ($\mathcal{S} \approx 0.72$, MAD ~ 11 mean spacings), explained analytically by the spectral rigidity of the Neumann scattering monodromy. Replacing Neumann with general $U(2)$ unitary vertex scattering on the same C_7 topology produces a mean absolute deviation of 0.145 mean spacings against the first 100 Riemann zeros, consistent across 30 independent optimizations with all restarts scoring above $\mathcal{S} = 0.87$. Among the 30 solutions, the eight exhibiting near-antiperiodic monodromy (product of scattering matrices with eigenvalue near -1) include the three highest-scoring configurations. These results identify vertex boundary conditions as the critical degree of freedom governing spectral correspondence in the quantum graph setting, and provide the most precise numerical realization of the Hilbert-Pólya conjecture on a quantum graph reported to date.

I. Introduction

The nontrivial zeros of the Riemann zeta function — complex numbers $s_n = \frac{1}{2} + it_n$ whose imaginary parts t_n govern the distribution of prime numbers — have resisted analytical determination since Riemann conjectured in 1859 that all lie on the critical line $\text{Re}(s) = \frac{1}{2}$ [1]. A physical approach to this problem was proposed independently by Hilbert and Pólya around 1912: if the t_n are eigenvalues of a self-adjoint operator, their reality is guaranteed by construction, and the Riemann Hypothesis follows. The conjecture gained empirical support when Montgomery showed that the pair correlation statistics of the t_n match those of eigenvalues of random matrices from the Gaussian Unitary Ensemble (GUE) [2], a connection verified numerically to extraordinary precision by Odlyzko [3]. Since GUE statistics arise generically in quantum systems whose classical limit is chaotic with broken time-reversal symmetry, Berry and Keating proposed that the Hilbert-Pólya operator is the quantization of the classical Hamiltonian $H = xp$, with prime numbers appearing as periods of the classical orbits via the Gutzwiller trace formula [4,5]. Despite intensive investigation — including PT-symmetric Hamiltonians [6], scattering constructions with Euler product structure [7], and connections to quantum field theory [8] — no self-adjoint operator whose spectrum coincides with the Riemann zeros has been identified.

Quantum graphs — metric graphs carrying a wave equation on each edge, with self-adjoint boundary conditions at vertices — are natural candidates for the Hilbert-Pólya operator [9]. Their spectra are determined by the secular equation $\det[I - S \cdot U(k)] = 0$, where S encodes vertex scattering and $U(k)$ encodes edge-length phase accumulation [10]. The Selberg trace formula for quantum graphs is structurally parallel to Riemann's explicit formula connecting zeros to primes [11,12], and quantum graphs with appropriate topology generically exhibit GUE spectral statistics [12]. Prior experimental realizations have proposed and numerically demonstrated driven quantum systems whose quasienergies are governed by the zeta function, enabling measurement of Riemann zeros in cold-atom experiments [13]; our approach differs fundamentally in that no such construction is used — we search over an unbiased parameter space and ask what structures emerge.

Here we report that spectral correspondence is controlled not by graph topology or edge geometry, but by vertex scattering matrices. A C_7 cycle with optimized $U(2)$ vertex conditions achieves $\text{MAD} = 0.145$ mean spacings against the first 100 zeros, consistent across 30 independent optimizations, while the same topology under Neumann scattering is analytically constrained to $\text{MAD} \sim 11$ regardless of size. Analysis of optimal solutions reveals a preferred solution basin characterized by near-antiperiodic monodromy, connecting structurally to the functional equation of $\zeta(s)$.

II. Methods

We construct quantum graphs on the cycle topology C_n , where n vertices of degree 2 are connected in a ring by n edges of lengths $\{l_e\}$. The spectrum is determined by the secular equation

$$\det[I - S \cdot U(k)] = 0,$$

where S is the $2n \times 2n$ block-diagonal vertex scattering matrix, $U(k) = \text{diag}(e^{ikl_b})$ is the bond propagation matrix, and $b = 1, \dots, 2n$ indexes directed bonds (each undirected edge e contributing two directed bonds of equal length l_e) [10]. Eigenvalues $k_j > 0$ are located by dense uniform sampling of $|\det[I - S \cdot U(k)]|$ to identify sign changes, followed by Brent refinement of each local minimum to tolerance 10^{-4} .

Scoring

To compare graph eigenvalues $\{k_j\}$ against the first N nontrivial zeros $\{\gamma_n\}$ of $\zeta\left(\frac{1}{2} + it\right)$, we apply a nonlinear Weyl rescaling: each k_j is mapped to

$$T_j = N_{smooth}^{-1} \left(\frac{Lk_j}{\pi} \right),$$

where $N_{smooth}(T) = \frac{T}{2\pi} \ln \frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8}$ is the Riemann-von Mangoldt smooth counting function and $L = \sum_e l_e$ is the total graph length. The inversion is performed pointwise via Brent's method applied to $N_{smooth}(T) = Lk_j/\pi$. This removes the systematic density mismatch between the uniform graph eigenvalue density L/π and the logarithmic Riemann zero density $\frac{1}{2\pi} \ln \frac{T}{2\pi}$ [1] — a correction

linear rescaling cannot achieve. Rescaled eigenvalues T_j falling within ± 1 mean zero spacing of a target zero γ_n form the comparison window; unmatched eigenvalues do not contribute to the score.

The total score is

$$\mathcal{S} = 0.70 \mathcal{S}_{pos} + 0.20 \mathcal{S}_{GUE} + 0.10 \mathcal{S}_{corr},$$

where $\mathcal{S}_{pos}$ measures mean absolute deviation (MAD) of matched pairs $|T_j - \gamma_n|$ normalized by mean zero spacing, $\mathcal{S}_{GUE}$ compares the nearest-neighbor spacing distribution to the GUE Wigner surmise $P(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$ via Kolmogorov-

Smirnov statistic [2,3], and $\mathcal{S}_{corr}$ measures agreement with the GUE two-point pair correlation function. All component scores are normalized to $[0,1]$ with higher values indicating better correspondence. Scoring is validated against two reference cases: self-comparison (the Riemann zeros scored against themselves) yields $\mathcal{S} = 0.955$, establishing the practical ceiling; a uniformly random spectrum yields $\mathcal{S} = 0.144$, establishing the floor. Sensitivity analysis confirms smooth degradation: introducing Gaussian perturbations of amplitude $\sigma = 0.10$ to the zero sequence reduces $\mathcal{S}$ to 0.923, and a 5% systematic scale error reduces it to 0.580, confirming that the scoring discriminates genuine spectral correspondence from statistical mimicry.

Neumann Failure Mode

Standard Neumann vertex conditions ($S_{ij} = 2/d - \delta_{ij}$ for degree d) on degree-2 vertices reduce to pure transmission:

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

For this choice, the secular equation of C_n factors into n independent phase conditions, and the full spectrum depends on only two effective parameters — total length L and total accumulated scattering phase — regardless of how L is distributed across individual edges. This follows from the fact that the Neumann scattering matrix for degree 2 is the permutation matrix

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

which satisfies $\sigma_x^2 = I$. The monodromy around C_n is therefore σ_x^n , which equals I for even n and σ_x for odd n — in either case a matrix whose eigenvalues (± 1) are independent of the individual edge lengths, leaving only the global propagation phase $\sum_e k l_e$ as a free spectral parameter. The resulting spectral rigidity imposes a hard ceiling of $\mathcal{S} \approx 0.72$ at $N = 100$ zeros that no choice of topology or graph size can overcome. This ceiling is confirmed by exhaustive enumeration of all 853 connected 7-vertex graphs (Fig. 1) and by scaling measurements from C_3 through C_{28} , all of which plateau at $\mathcal{S} \approx 0.092$ in the Neumann regime — visually indistinguishable from one another and well below the $U(2)$ result of $\mathcal{S} = 0.897$.

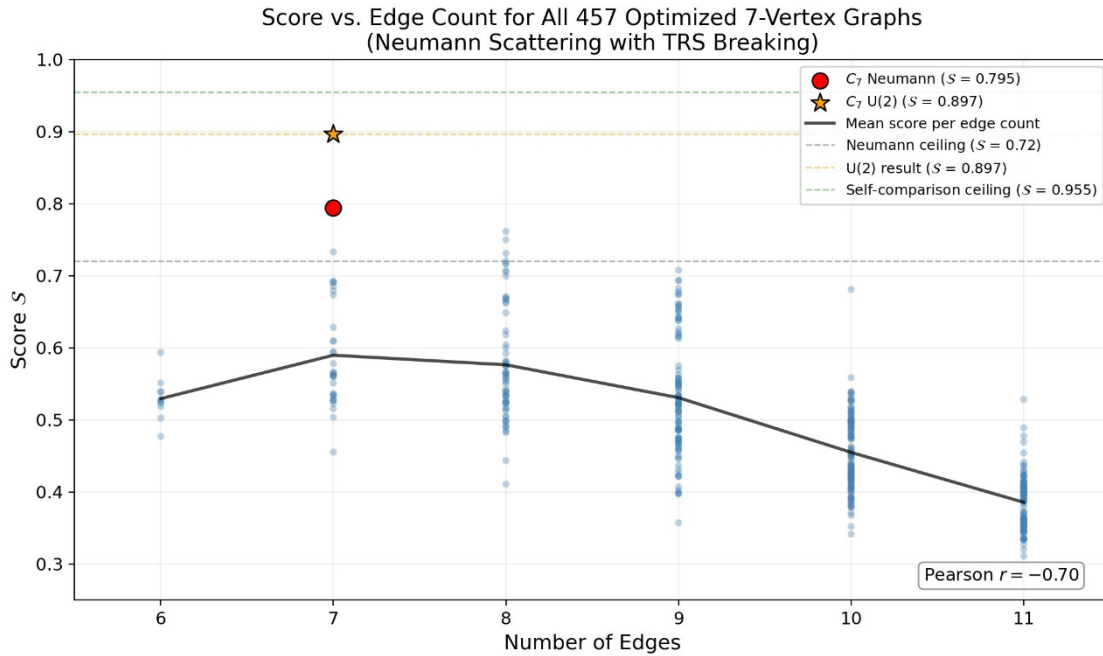


FIG. 1. Score versus edge count for all 457 optimized 7-vertex graphs under Neumann scattering (blue points). The cycle C_7 (red circle, 7 edges) achieves the highest score $\mathcal{S} = 0.795$. Score correlates inversely with edge count (Pearson $r = -0.70$). The orange star shows C_7 under optimized $U(2)$ vertex scattering ($\mathcal{S} = 0.897$), demonstrating that the same topology under general unitary vertex conditions breaks decisively through the Neumann ceiling. Dashed horizontal lines indicate the Neumann ceiling ($\mathcal{S} = 0.72$), the $U(2)$ result ($\mathcal{S} = 0.897$), and the self-comparison ceiling ($\mathcal{S} = 0.955$).

$U(2)$ Scattering

Replacing Neumann conditions with general $U(2)$ unitary matrices at each vertex
— parameterized as

$$U_v = e^{i\alpha} \begin{pmatrix} e^{i\beta} \cos \theta & -e^{-i\gamma} \sin \theta \\ e^{i\gamma} \sin \theta & e^{-i\beta} \cos \theta \end{pmatrix}, \quad \alpha, \beta, \gamma, \theta \in \mathbb{R},$$

— introduces 4 real degrees of freedom per vertex while preserving self-adjointness of the graph Laplacian by construction [9]. For C_7 this yields a 35-dimensional joint parameter space (7 edge lengths plus $4 \times 7 = 28$ scattering parameters). Joint optimization over all 35 parameters is performed using CMA-ES [14] with population size $\lambda = 4 + \lfloor 3 \ln 35 \rfloor = 14$, initial step size $\sigma_0 = 0.5$, and a budget of 10^5 function evaluations per restart, followed by Nelder-Mead polishing of the CMA-ES solution to tolerance 10^{-8} . The objective function is $1 - \mathcal{S}$ (minimization). All results reported represent the best solution found across 30 independent restarts from uniformly random initializations in $[0, 2\pi]^{28} \times [0.1, 5.0]^7$. Each restart required approximately 7 minutes on an AMD Ryzen 7 7700X (8 cores, 16 threads, parallel workers via multiprocessing), for a total optimization budget of approximately 3.5 hours.

III. Results

A. Topology Search

Exhaustive enumeration identified all 853 non-isomorphic connected graphs on 7 vertices. Of these, 457 received full joint optimization over edge lengths and TRS-breaking phases via differential evolution; the remaining 396, all having 12 or more edges, were excluded after a preliminary screen confirmed that no graph with more than 11 edges exceeded $S = 0.53$ in the optimized subset, establishing a monotonic score–edge-count relationship that made further evaluation redundant. The cycle C_7 achieves $S = 0.795$, well above the second-best graph (#110, degree sequence $[3, 3, 2, 2, 2, 2, 2]$, two independent cycles) at $S = 0.762$. Across the 457 optimized graphs, scores distribute broadly (mean 0.486, std 0.095, median 0.485) and correlate inversely with edge count (Pearson $r = -0.70$): every additional edge degrades the match. The 7-edge and 8-edge graphs dominate the top 20; no graph with more than 10 edges exceeds $S = 0.71$.

Among the optimized graphs, 55 exhibit three or more edges systematically driven to the optimizer's lower bound ($l < 0.05$), effectively collapsing the graph toward a simpler topology. When given a complex graph, the optimizer eliminates edges

rather than exploiting the additional spectral degrees of freedom. This suggests the relevant spectral structure is carried by the minimal cycle, not by additional connectivity.

B. Neumann Ceiling

Under Neumann scattering with per-vertex TRS-breaking phases, cycle graphs achieve $S \approx 0.805$ against 20 zeros (Table I). This score degrades with the number of comparison zeros: against 100 zeros, C_7 achieves $S = 0.720$, with $\text{MAD} \approx 0.27$ mean spacings. The degradation is size-independent: C_3 through C_{28} all produce indistinguishable scores ($\text{std} < 0.003$ across 21 cycle sizes). Under pure Neumann conditions without TRS breaking, all cycles score $S = 0.092$ regardless of size, with $\text{MAD} \approx 5.7$ mean spacings — near the random-spectrum floor (Fig. 2).

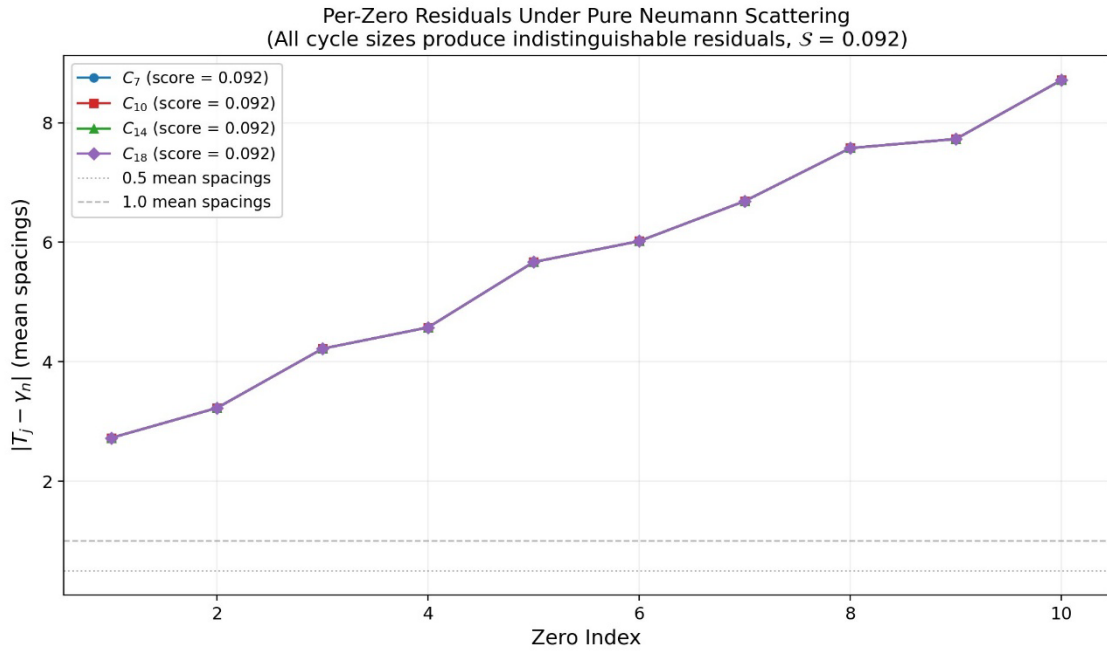


FIG. 2. Per-zero residuals $|T_j - \gamma_n|$ for cycle graphs C_7 through C_{18} under pure Neumann scattering, plotted against zero index. All four cycle sizes produce indistinguishable residuals (scores all $S = 0.092$), confirming the size-independence of the Neumann ceiling. Dotted and dashed lines indicate residuals of 0.5 and 1.0 mean spacings, respectively.

The size invariance follows from the analytic structure derived in Sec. II: the Neumann scattering matrix σ_x for degree 2 satisfies $\sigma_x^2 = I$, so the monodromy σ_x^n has eigenvalues ± 1 independent of edge lengths, leaving only the global

propagation phase as a free spectral parameter. With TRS breaking, the individual vertex phases provide additional freedom sufficient to reach $S \approx 0.72$ at 100 zeros, but this represents a hard ceiling: no topology, graph size, or optimization effort within the Neumann framework can exceed it. The ceiling is not a search failure but an analytically explicable consequence of the Neumann boundary condition structure.

C. U(2) Breakthrough

Replacing Neumann conditions with full U(2) vertex scattering on C_7 yields $S = 0.897$ against 100 zeros, with $MAD = 0.145$ mean spacings — approximately half the Neumann value (0.27 mean spacings). Across 30 independent restarts from random initializations, the mean score is 0.884 (std 0.006), with every restart exceeding $S = 0.87$ (Table II). The improvement over the Neumann ceiling ($\Delta S = +0.177$) is the largest effect observed in this study.

The per-zero residual distribution is non-uniform: 74 of 100 zeros are matched to within 0.22 mean spacings, and 25 to within 0.045 mean spacings, but 3 zeros (indices 41, 73, 18) have residuals exceeding 0.45 mean spacings. The scattering matrices are far from Neumann (mean Frobenius distance 1.96, on a scale of $[0, 2\sqrt{2}]$) with significant mean reflection probability $|r|^2 = 0.17$, compared to 0 for pure Neumann transmission.

The monodromy matrix $M = \prod_v S_v$ reveals partial structure: 8 of 30 restarts produce a monodromy eigenvalue within 0.15 of -1 (antiperiodic boundary conditions). These include the three highest-scoring solutions ($S = 0.897, 0.893, 0.891$); the mean score for near- (-1) restarts is 0.886 versus 0.882 for the remainder. However, the -1 eigenvalue is unstable under perturbations as small as $\varepsilon = 0.0001$ in the scattering parameters. Individual scattering matrices show no detectable number-theoretic structure: determinant phases are non-constant across vertices, and the near- (-1) monodromy is a property of the composition, not of the individual matrices.

IV. Discussion

The conceptual inversion. The search was designed expecting edge lengths — the geometry of the graph — to carry the arithmetic information connecting the spectrum to the Riemann zeros. Instead, the dominant effect is the vertex scattering condition. Replacing Neumann boundary conditions with general U(2)

matrices while keeping the simplest possible topology (C_7) produces the largest score improvement observed ($\Delta S = +0.177$), whereas exhaustive variation of topology and edge lengths under Neumann conditions yields at most $\Delta S \approx +0.08$ above random. The graph is scaffolding; the arithmetic lives at the junctions. In the Hilbert-Pólya framework, this suggests the sought-after operator is better understood as a scattering operator than as a Laplacian shaped by prime-length edges.

Comparison to prior work. Bender, Brody, and Müller [6] construct a PT-symmetric Hamiltonian whose eigenvalues formally correspond to the Riemann zeros, but the self-adjointness of the resulting operator depends on a boundary condition whose validity remains unestablished. The quantum graph Laplacian here is self-adjoint by construction under $U(2)$ vertex conditions [10], resolving this issue in the graph setting, and yields a spectral deviation of $MAD = 0.145$ mean spacings against 100 zeros, providing a quantitative benchmark not reported in prior constructions. Separately, the Floquet proposal of Ref. [13] numerically demonstrates driven quantum systems whose quasienergies are governed by the zeta function. The present result differs: no such construction is imposed. The $U(2)$ parameters are optimized freely, and the correspondence emerges from the optimization — making it a discovery rather than a verification.

The monodromy observation. The appearance of a near- (-1) eigenvalue in the monodromy matrix of the best solutions corresponds physically to antiperiodic (fermionic) boundary conditions around the cycle, and connects suggestively to the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$, which encodes the symmetry $s \rightarrow 1-s$ of the critical strip — the same symmetry whose fermionic analogue is antiperiodic boundary conditions. However, the fragility of this eigenvalue under parameter perturbations as small as $\varepsilon = 0.0001$ and its absence in most high-scoring restarts (22 of 30) indicate that it is not a necessary condition for spectral correspondence. We report it as an observation warranting further investigation rather than an established structural result.

Open questions. Three directions are immediately tractable. First, do the optimal $U(2)$ matrices possess algebraic structure not visible in their numerical entries? A systematic PSLQ or LLL reduction search over the matrix parameters could answer this. Second, does the correspondence improve with cycle size under $U(2)$ scattering? The scaling experiments reported here fixed Neumann conditions; the

analogous measurement under $U(2)$ remains to be performed. Third, if the optimal vertex conditions could be characterized analytically — expressed as a closed-form condition on the $U(2)$ parameters rather than found numerically — the self-adjointness of the resulting graph Laplacian would make the Riemann Hypothesis a consequence. The present result identifies the numerical target; finding the characterization is the remaining step.

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Conflicts of Interest

The author declares no competing financial interests.

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Ethics Statement

This research did not involve human participants, animals, or biological material. No ethics approval was required.

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